Investigating the Double Descent Phenomenon

Factorization Models on Sparse E-Commerce Transaction Data

Denise Brancaleone Matías Cerda Gianluca Pasquinelli Marco Restuccia

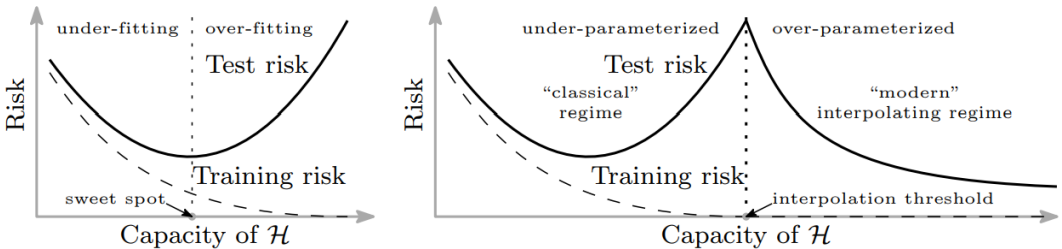
M.Sc. in Data Science
Statistical Machine Learning
Sapienza University of Rome

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Can Double Descent Emerge from Sparse Purchase Data?

- Real-world recommender systems are often built from **sparse implicit feedback**, where purchases indicate interest but missing interactions are not true dislikes.
- Matrix factorization models represent users and items in a latent space whose dimension k directly controls model capacity.
- Classical learning theory suggests that increasing complexity eventually leads to overfitting.
- Modern overparameterized models may instead improve again after a critical region, producing a **double-descent** curve.
- **Research question:** Does recommendation performance exhibit double descent as k increases in sparse implicit-feedback data?

Beyond the Classical Bias–Variance Trade-Off



Source: Belkin et al. (2019), Figure 1.

- Classical theory predicts an optimal intermediate complexity.
- Near the interpolation threshold, test error may reach a peak.
- Beyond this point, larger models can improve again.

Dataset: A Large but Extremely Sparse Purchase Matrix

- Shopify transactions from *Fondo de Cultura Económica Chile*, covering October 2019–December 2025.
- Binary utility matrix $X \in \{0,1\}^{n \times m}$: $X_{ui} = 1$ if user u purchased item i, and $X_{ui} = 0$ otherwise. Zero denotes an unobserved interaction, not an explicit dislike.
- 23,605 users, 4,405 items, and 136,832 positive user–item interactions.
- Extremely sparse matrix: density 0.13% and sparsity 99.86%.
- Missing interactions are not true negatives, since customers may purchase through other channels or simply not have been exposed to the work.

Models: Three Ways to Learn from Sparse Feedback

- **Non-Negative Matrix Factorization (NMF):** reconstructs the binary interaction matrix using a squared-error loss evaluated only on a fixed sampled support Ω, composed of observed purchases and sampled unobserved pairs. User and item factors are constrained to be non-negative.
- **Weighted Matrix Factorization / iALS:** optimizes a weighted squared-error objective over the entire user–item matrix. Observed purchases receive higher confidence, while every unobserved cell still contributes to the loss with a lower weight.
- **Logistic Matrix Factorization:** models the probability of a purchase through a logistic function and is trained with binary cross-entropy. Positive interactions are combined with sampled unobserved pairs used as negative examples.

Common Experimental Protocol

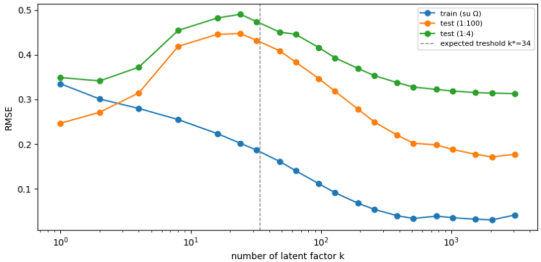
- Common notation: $P \in \mathbb{R}^{n \times k}$ and $Q \in \mathbb{R}^{m \times k}$ denote the user and item factor matrices, with latent vectors $p_u, q_i \in \mathbb{R}^k$.
- Random leave-one-out split at user level; cold test items removed.
- Fixed split and fixed random seeds across each capacity sweep.
- Activity thresholds $\tau \in \{3, 5, 10, 20, 25\}$.
- Pointwise evaluation with RMSE; full-catalog ranking with NDCG@K.
- For NMF and Logistic MF, $\Omega = \mathcal{P} \cup \mathcal{N}$, where $\mathcal{P}$ denotes training positives and $\mathcal{N}$ sampled unobserved pairs, with $|\mathcal{N}| = 4|\mathcal{P}|$.

NMF: Model and Training Support

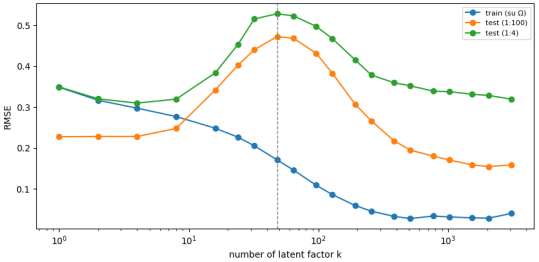
$$\min_{P,Q\geq 0} \sum_{(u,i)\in\Omega} \left(X_{ui} - p_u^\top q_i\right)^2 + \lambda \left(\|P\|_F^2 + \|Q\|_F^2\right)$$

- Ω: observed purchases plus sampled negatives.
- Cells outside Ω remain unknown and do not enter the loss.
- k controls the transition from low-rank to overparameterized models.

NMF: Activity Restores the First Descent

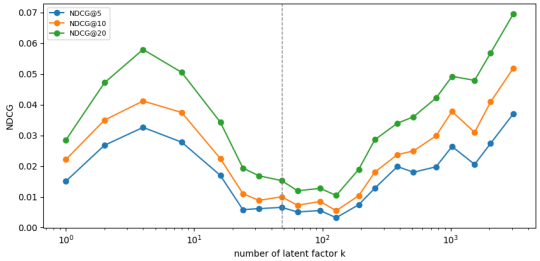


$\tau = 3,\ \lambda = 0.06$

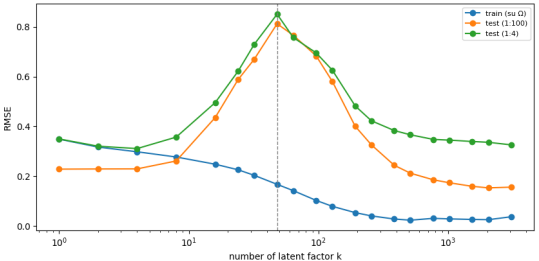


$\tau = 20,\ \lambda = 0.06$

NMF: Ranking and Regularization



NDCG, $\tau = 20$, $\lambda = 0.06$



RMSE, $\tau = 20$, $\lambda = 0.01$

- Higher user activity makes the underparameterized regime more visible.
- Weaker regularization amplifies the critical peak without changing its location.

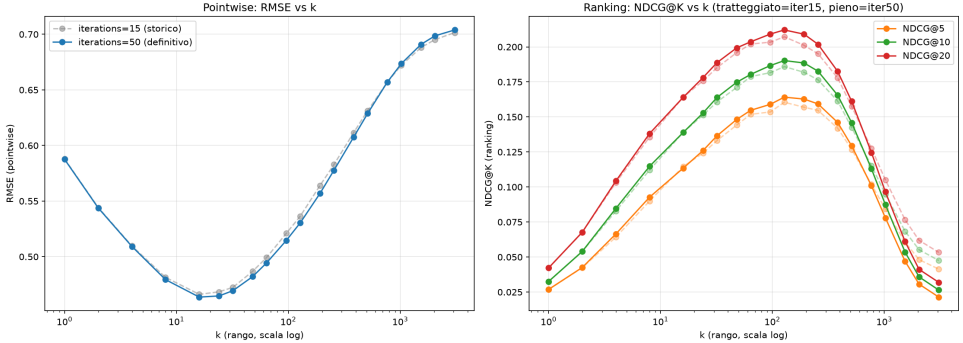
Weighted MF / iALS

$$\min_{P,Q} \quad \sum_{u=1}^{n} \sum_{i=1}^{m} C_{ui} \left(X_{ui} - p_u^\top q_i \right)^2 + \lambda \left(\|P\|_F^2 + \|Q\|_F^2 \right),$$

$$\text{where} \quad C_{ui} = 1 + \alpha X_{ui}.$$

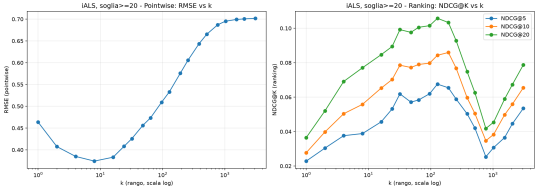
- Every user–item cell contributes to training.
- Positives and unobserved cells differ through confidence C_{ui}.
- No non-negativity constraint and no explicit negative sampling.

iALS: Capacity and Training-Time Behavior

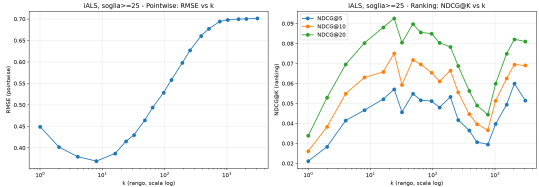


- At $\tau = 3$, NDCG peaks near $k = 128$ and then deteriorates, while RMSE worsens for larger models.
- Additional iterations help medium k but worsen very large models.

iALS: Recovery Appears for Denser Populations



$\tau = 20$



$\tau = 25$

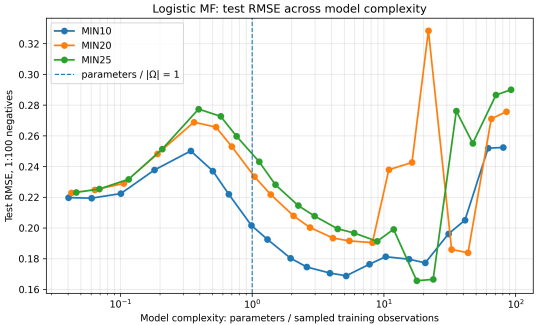
- A sustained NDCG recovery emerges only for the smaller, denser populations.
- The recovery is much weaker or absent in RMSE.

Logistic Matrix Factorization

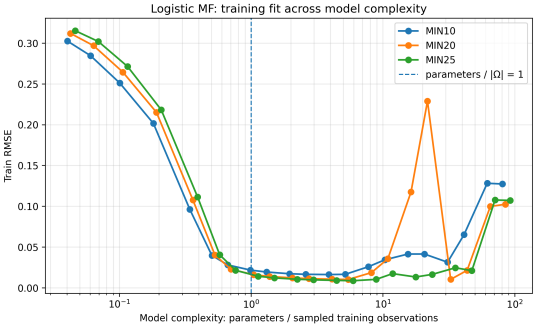
$$\widehat{X}_{ui} = \Pr(X_{ui} = 1 \mid u, i) = \sigma\Big(\mu + b_u + b_i + p_u^\top q_i\Big)$$

- $\widehat{X}_{ui} \in (0, 1)$ is the predicted purchase probability.
- Binary cross-entropy objective.
- Four sampled negatives per observed purchase.
- User and item biases separate global popularity from latent interactions.

Initial Logistic MF Sweep



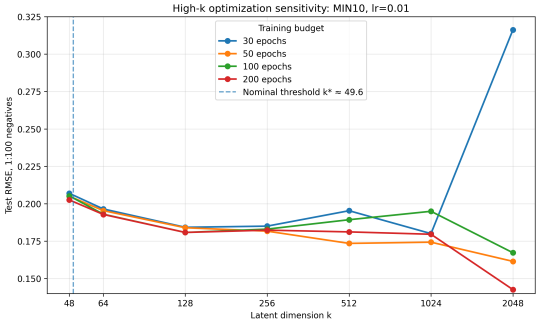
Test RMSE



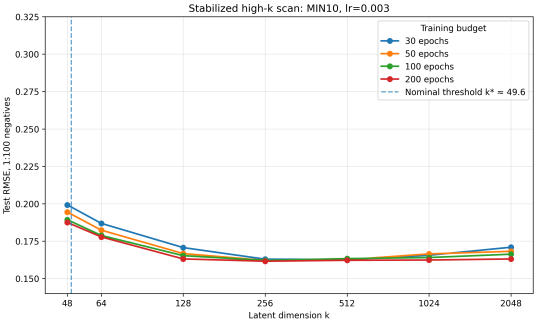
Train RMSE

- A second high-k peak appears in both train and test RMSE.
- This points to optimization instability rather than pure overfitting.

Optimization Diagnostics



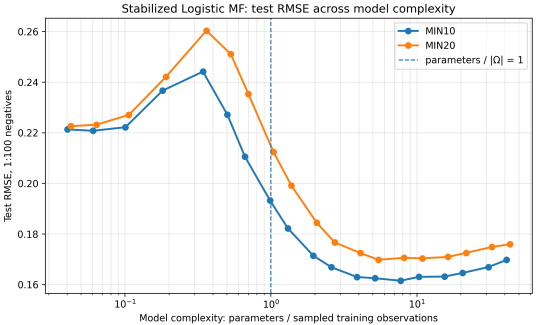
Test RMSE, lr=0.01



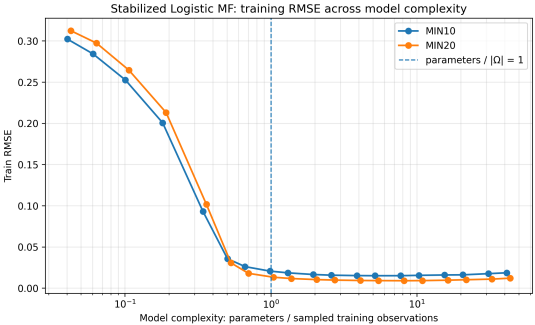
Test RMSE, lr=0.003

- With $lr = 0.01$, the 30-epoch model shows a sharp high-k spike.
- Increasing the training budget or reducing the learning rate removes this instability.
- The evidence therefore supports interpreting the second high-k peak as optimization-driven instability rather than robust double descent.

Stabilized Logistic MF Sweep



Stabilized test RMSE



Stabilized train RMSE

- Lower learning rate and gradient clipping remove the high-capacity spike.
- The intermediate peak and second-descent-like recovery remain.

Cross-Model Comparison

	NMF	iALS	Logistic MF
Training support	Sampled	Full matrix	Sampled
Loss	Squared	Weighted squared	BCE
Non-negativity	Yes	No	No
Main DD evidence	RMSE + NDCG	NDCG, high τ	RMSE
Main sensitivity	τ, λ	τ, iterations	τ, optimization

- Double descent is not a universal property of matrix factorization.
- Its visibility depends on training support, population density, metric, and optimization.

Main Conclusions

- Double-descent-like behavior survives in sparse implicit-feedback recommendation, but conditionally.
- NMF provides the clearest and most consistent evidence across thresholds.
- iALS shows recovery mainly for denser user populations and primarily in ranking quality.
- Logistic MF requires optimization diagnostics to separate capacity effects from instability.
- RMSE and NDCG reveal different aspects of generalization and must be interpreted jointly.

Questions? Key takeaways from the three models

NMF

Clearest evidence of a double-descent-like pattern across RMSE and NDCG.

iALS

Recovery appears mainly for denser user populations and ranking metrics.

Logistic MF

Optimization diagnostics are essential to separate capacity effects from instability.

Double descent appears conditionally, not universally.